

DEPARTMENT:PDI

LAB 04: File_IO

Course: PDI

Group: I2-IP-Group AIECS

Submitted by:

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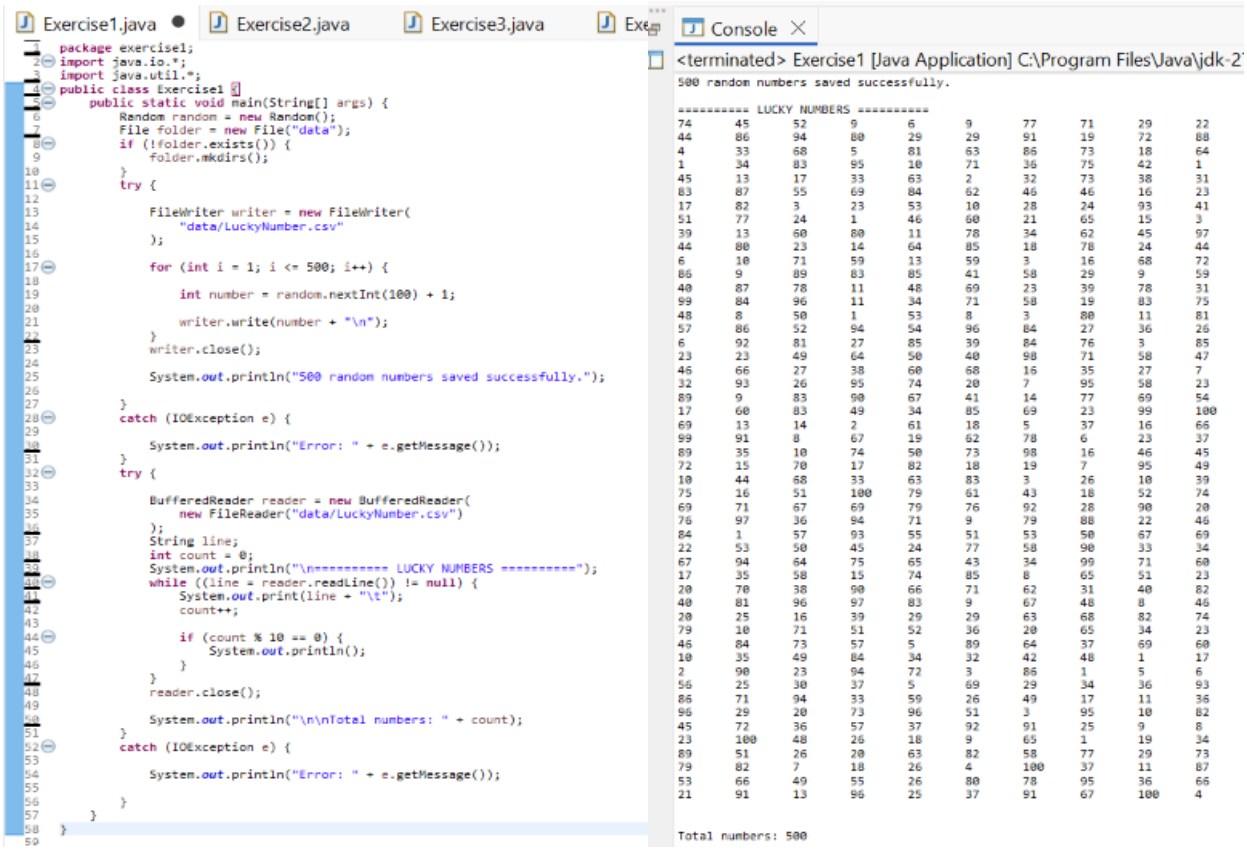
Group: I2-IP-Group

Submitted to: Channa Bou

Academic Year: 2026–2027

Institute of Technology of Cambodia

EXERCISE1:



Exercise1.java ● Exercise2.java Exercise3.java Exe

```java
package exercise1;
import java.io.*;
import java.util.*;
public class Exercise1 {
    public static void main(String[] args) {
        Random random = new Random();
        File folder = new File("data");
        if (!folder.exists()) {
            folder.mkdirs();
        }
        try {

            FileWriter writer = new FileWriter(
                "data/LuckyNumber.csv"
            );

            for (int i = 1; i <= 500; i++) {

                int number = random.nextInt(100) + 1;

                writer.write(number + "\n");
            }
            writer.close();

            System.out.println("500 random numbers saved successfully.");

        }
        catch (IOException e) {

            System.out.println("Error: " + e.getMessage());
        }
        try {

            BufferedReader reader = new BufferedReader(
                new FileReader("data/LuckyNumber.csv")
            );
            String line;
            int count = 0;
            System.out.println("\n========== LUCKY NUMBERS ==========");
            while ((line = reader.readLine()) != null) {
                System.out.print(line + "\t");
                count++;

                if (count % 10 == 0) {
                    System.out.println();
                }
            }
            reader.close();

            System.out.println("\n\nTotal numbers: " + count);
        }
        catch (IOException e) {

            System.out.println("Error: " + e.getMessage());

        }
    }
}
```

Console

<terminated> Exercise1 [Java Application] C:\Program Files\Java\jdk-2

```
500 random numbers saved successfully.

========== LUCKY NUMBERS ==========
74      45      52      9       6       9       77      71      29      22
44      86      94      80      29      29      91      19      72      88
4       33      68      5       81      63      86      73      18      64
1       34      83      95      10      71      36      75      42      1
45      13      17      33      63      2       32      73      38      31
83      87      55      69      84      62      46      46      16      23
17      82      3       23      53      10      28      24      93      41
51      77      24      1       46      60      21      65      15      3
39      13      60      80      11      78      34      62      45      97
44      80      23      14      64      85      18      78      24      44
6       10      71      59      13      59      3       16      68      72
86      9       89      83      85      41      58      29      9       59
40      87      78      11      48      69      23      39      78      31
99      84      96      11      34      71      58      19      83      75
48      8       50      1       53      8       3       80      11      81
57      86      52      94      54      96      84      27      36      26
6       92      81      27      85      39      84      76      3       85
23      23      49      64      50      40      98      71      58      47
46      66      27      38      60      68      16      35      27      7
32      93      26      95      74      20      7       95      58      23
89      9       83      90      67      41      14      77      69      54
17      60      83      49      34      85      69      23      99      100
69      13      14      2       61      18      5       37      16      66
99      91      8       67      19      62      78      6       23      37
89      35      10      74      50      73      98      16      46      45
72      15      70      17      82      18      19      7       95      49
10      44      68      33      63      83      3       26      10      39
75      16      51      100     79      61      43      18      52      74
69      71      67      69      79      76      92      28      90      20
76      97      36      94      71      9       79      88      22      46
84      1       57      93      55      51      53      50      67      69
22      53      50      45      24      77      58      90      33      34
67      94      64      75      65      43      34      99      71      60
17      35      58      15      74      85      8       65      51      23
20      70      38      90      66      71      62      31      40      82
40      81      96      97      83      9       67      48      8       46
20      25      16      39      29      29      63      68      82      74
79      10      71      51      52      36      20      65      34      23
46      84      73      57      5       89      64      37      69      60
10      35      49      84      34      32      42      48      1       17
2       90      23      94      72      3       86      1       5       6
56      25      30      37      5       69      29      34      36      93
86      71      94      33      59      26      49      17      11      36
96      29      20      73      96      51      3       95      10      82
45      72      36      57      37      92      91      25      9       8
23      100     48      26      18      9       65      1       19      34
89      51      26      20      63      82      58      77      29      73
79      82      7       18      26      4       100     37      11      87
53      66      49      55      26      80      78      95      36      66
21      91      13      96      25      37      91      67      100     4


Total numbers: 500
```

EXERCISE2:

```java
package exercise2;
import java.io.*;
import java.util.*;
public class Exercise2 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        Random random = new Random();
        System.out.print("Enter number of files: ");
        int n = input.nextInt();
        File folder = new File("data/Ex2LuckyNumber");
        if (!folder.exists()) {
            folder.mkdirs();
        }
        for (int i = 1; i <= n; i++) {

            try {
                FileWriter writer = new FileWriter(
                    "data/Ex2LuckyNumber/file" + i + ".csv"
                );
                for (int j = 1; j <= 10; j++) {
                    int number = random.nextInt(100) + 1;
                    writer.write(number + "\n");
                }
                writer.close();
                System.out.println("file" + i + ".csv created.");
            }
            catch (IOException e) {
                System.out.println(e.getMessage());
            }
        }
        ArrayList<Integer> numbers = new ArrayList<>();
        for (int i = 1; i <= n; i++) {
            try {

                BufferedReader reader = new BufferedReader(
                    new FileReader(
                        "data/Ex2LuckyNumber/file" + i + ".csv"
                    )
                );

                String line;
                while ((line = reader.readLine()) != null) {

                    numbers.add(Integer.parseInt(line));
                }

                reader.close();
            }
            catch (IOException e) {
                System.out.println(e.getMessage());
            }
        }
        if (!numbers.isEmpty()) {
            int max = Collections.max(numbers);
            int min = Collections.min(numbers);
            System.out.println("\n========== RESULTS ==========");
            System.out.println("All numbers: " + numbers);
            System.out.println("Total numbers: " + numbers.size());
            System.out.println("Maximum number: " + max);
            System.out.println("Minimum number: " + min);
        }
        input.close();
    }
}
```

OUTPUT:

```
Enter number of files: 7
file1.csv created.
file2.csv created.
file3.csv created.
file4.csv created.
file5.csv created.
file6.csv created.
file7.csv created.

========== RESULTS ==========
All numbers: [20, 78, 66, 3, 100, 26, 6, 96, 48, 38, 58, 94, 70, 97, 6, 70, 10, 95, 11, 98, 20, 36, 92, 2, 8, 77, 81, 6, 33, 84, 24, 47, 38, 82, 64, 33, 31, 86, 41, 73, 54, 72, 10, 93, 74, 71, 89, 13, 6, 56, 52, 73, 94, 93, 76, 57, 66, 47, 27, 32, 76, 5, 9, 38, 81,
22, 1, 97, 81, 80]
Total numbers: 70
Maximum number: 100
Minimum number: 1
```

EXERCISE3:

Exercise1.java | Exercise2.java | Exercise3.java

```java
package exercise3;
import java.io.*;

public class Exercise3 {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
        File folder = new File("data");

        if (!folder.exists()) {
            folder.mkdirs();
        }


        try {

            FileWriter writer = new FileWriter(
                "data/StudentData.csv"
            );


            writer.write("Name,ID,Gender\n");


            for (int i = 1; i <= 3; i++) {

                System.out.println("\nStudent " + i);

                System.out.print("Enter name: ");
                String name = input.nextLine();

                System.out.print("Enter ID: ");
                String id = input.nextLine();

                System.out.print("Enter gender: ");
                String gender = input.nextLine();


                writer.write(name + "," + id + "," + gender + "\n");
            }

            writer.close();

            System.out.println("\nStudent data saved successfully.");
        }
        catch (IOException e) {

            System.out.println("Error: " + e.getMessage());
        }
        input.close();
    }
}
```

OUTPUT:

Console

<terminated> Exercise3 [Java Applicat

```
Student 1
Enter name: fvvf
Enter ID: 43534
Enter gender: vfb

Student 2
Enter name: vtrbtr
Enter ID: 5353
Enter gender: fbgff

Student 3
Enter name: tbtr
Enter ID: 4453
Enter gender: trr

Student data saved successfully.
```

EXERCISE4:

```java
package exercise4;
import java.io.*;
import java.util.*;
public class Exercise4 {
    public static void main(String[] args) {
        ArrayList<String> students = new ArrayList<>();


        try {

            BufferedReader reader = new BufferedReader(
                new FileReader("data/StudentData.csv")
            );

            String line;
            reader.readLine();

            while ((line = reader.readLine()) != null) {

                students.add(line);

            }

            reader.close();

            System.out.println("Student data loaded successfully.");

        }
        catch (IOException e) {

            System.out.println("Error: " + e.getMessage());

        }
        System.out.println("\n========== STUDENT INFORMATION ==========");

        System.out.println("Name\t\tID\tGender");

        for (String student : students) {

            String[] data = student.split(",");

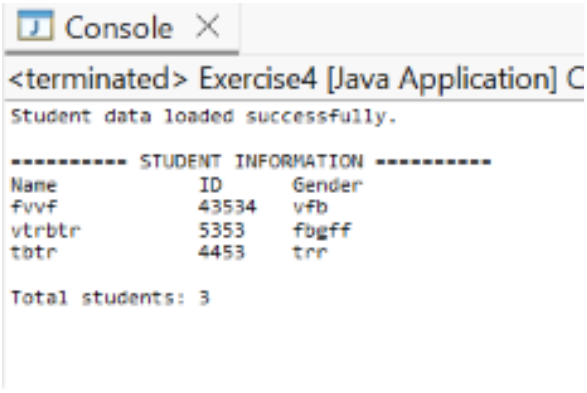
            String name = data[0].trim();
            String id = data[1].trim();
            String gender = data[2].trim();

            System.out.println(
                name + "\t\t" + id + "\t" + gender
            );
        }

        System.out.println("\nTotal students: " + students.size());

    }
}
```

OUTPUT:



EXERCISE5:

```java
package exercise5;
import java.io.*;
import java.util.*;
public class Exercise5 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        ArrayList<String> students = new ArrayList<>();
        try {
            BufferedReader reader = new BufferedReader(
                new FileReader("data/StudentData.csv")
            );
            String line;
            reader.readLine();
            while ((line = reader.readLine()) != null) {
                students.add(line);
            }
            reader.close();
            System.out.println("Existing student data loaded.");
        }
        catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
            input.close();
            return;
        }
        System.out.println("\n========== ADD 3 STUDENTS ==========");
        for (int i = 1; i <= 3; i++) {
            System.out.println("\nStudent " + i);
            System.out.print("Enter name: ");
            String name = input.nextLine();
            System.out.print("Enter ID: ");
            String id = input.nextLine();
            System.out.print("Enter gender: ");
            String gender = input.nextLine();
            students.add(name + "," + id + "," + gender);
        }
        Collections.sort(students, new Comparator<String>() {
            public int compare(String s1, String s2) {
                String name1 = s1.split(",")[0].trim();
                String name2 = s2.split(",")[0].trim();
                return name1.compareToIgnoreCase(name2);
            }
        });
        try {
            FileWriter writer = new FileWriter(
                "data/StudentData.csv"
            );
            writer.write("Name,ID,Gender\n");
            for (String student : students) {
                writer.write(student + "\n");
            }
            writer.close();
            System.out.println("\nStudent data updated successfully.");
        }
        catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
        System.out.println("\n========== SORTED STUDENT LIST ==========");
        for (String student : students) {
            String[] data = student.split(",");
            System.out.println(
                data[0] + "\t" + data[1] + "\t" + data[2]
            );
        }
        System.out.println("\nTotal students: " + students.size());
        input.close();
    }
}
```

OUTPUT:

```
Console
<terminated> Exercise5 [Java Application]
Existing student data loaded.

========== ADD 3 STUDENTS ==========

Student 1
Enter name: frg
Enter ID: 4345
Enter gender: rge

Student 2
Enter name: ergre
Enter ID: 5353
Enter gender: gr

Student 3
Enter name: eew
Enter ID: 3242
Enter gender: vr

Student data updated successfully.

========== SORTED STUDENT LIST ==========
eew     3242    vr
ergre   5353    gr
frg     4345    rge
fvvf    43534   vfb
tbtr    4453    trr
vtrbtr  5353    fbgff

Total students: 6
```

STUDENT FILE:

Exercise1.java Exercise2.java Exercise3.ja

```java
package common;
public class Student {
    private String name;
    private String id;
    private String gender;

    public Student(String name, String id, String gender) {
        this.name = name;
        this.id = id;
        this.gender = gender;
    }

    public String getName() {
        return name;
    }

    public String getId() {
        return id;
    }

    public String getGender() {
        return gender;
    }

    @Override
    public String toString() {
        return name + "," + id + "," + gender;
    }
}
```

EXERCISE6:

Exercise1.java Exercise2.java Exercise3.java E

```java
package exercise6;
import java.io.*;
import java.util.*;
public class Exercise6 {
    public static void readCSV() {
        int total = 0;
        int male = 0;
        int female = 0;
        try {

            BufferedReader reader =
                new BufferedReader(
                    new FileReader("data/CandidatePerformanceTest.csv")
                );

            String line;
            String header = reader.readLine();
            System.out.println(header);
            System.out.println("--------------------------------");
            while ((line = reader.readLine()) != null) {
                String[] data = line.split(",");
                String name = data[0].trim();
                String gender = data[1].trim();
                System.out.println(line);
                total++;
                if (gender.equalsIgnoreCase("Male")) {
                    male++;
                }
                else if (gender.equalsIgnoreCase("Female")) {
                    female++;
                }
            }
            reader.close();
            double malePercentage = (male * 100.0) / total;
            double femalePercentage = (female * 100.0) / total;
            System.out.println("\n========== SUMMARY ==========");
            System.out.println("Total candidates: " + total);
            System.out.println("Male candidates: " + male);
            System.out.println("Female candidates: " + female);
            System.out.printf("Male percentage: %.2f%%%n",
                malePercentage);
            System.out.printf("Female percentage: %.2f%%%n",
                femalePercentage);
        }
        catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }

    public static void main(String[] args) {
        readCSV();
    }
}
```

OUTPUT:

```
2496,female,group D,some college,completed,75,77,83,235,78.33
8870,female,group E,some high school,none,74,74,72,220,73.33
4748,female,group C,some college,none,62,69,69,200,66.67
5258,female,group D,master degree,none,92,100,100,292,97.33

========== SUMMARY ==========
Total candidates: 960
Male candidates: 467
Female candidates: 491
Male percentage: 48.65%
Female percentage: 51.15%
```

EXERCISE7:

```java
package exercise7;
import java.io.*;
import java.util.*;
public class Exercise7 {
    static ArrayList<String> groupA = new ArrayList<>();
    static ArrayList<String> groupB = new ArrayList<>();
    static ArrayList<String> groupC = new ArrayList<>();
    static ArrayList<String> groupD = new ArrayList<>();

    public static void readCSV() {
        try {
            BufferedReader br = new BufferedReader(
                new FileReader("data/CandidatePerformanceTest.csv"));

            String line;
            br.readLine();

            while ((line = br.readLine()) != null) {
                String group = line.split(",")[2].trim();

                if (group.contains("A")) groupA.add(line);
                else if (group.contains("B")) groupB.add(line);
                else if (group.contains("C")) groupC.add(line);
                else if (group.contains("D")) groupD.add(line);
            }

            br.close();
            System.out.println("CSV loaded successfully.");

        } catch (IOException e) {
            System.out.println(e.getMessage());
        }
    }


    public static void processGroup(String name,
                                    ArrayList<String> students) {
        int male = 0, female = 0;

        for (String student : students) {
            String gender = student.split(",")[1].trim();

            if (gender.equalsIgnoreCase("Male")) male++;
            else if (gender.equalsIgnoreCase("Female")) female++;
        }

        int total = students.size();
        double malePercent = total == 0 ? 0 : male * 100.0 / total;
        double femalePercent = total == 0 ? 0 : female * 100.0 / total;

        System.out.println("\n" + name);
        System.out.println("Total students: " + total);
        System.out.println("Male: " + male);
        System.out.println("Female: " + female);
        System.out.printf("Male percentage: %.2f%%\n", malePercent);
        System.out.printf("Female percentage: %.2f%%\n", femalePercent);

        try {
            File folder = new File("data/Exercise7");
            folder.mkdirs();

            String fileName = name.replace(" ", "").toLowerCase() + ".csv";
            FileWriter fw = new FileWriter(new File(folder, fileName));
            fw.write("Name,Gender,Group\n");

            for (String student : students) {
                fw.write(student + "\n");
            }
            fw.write("\nSUMMARY\n");
            fw.write("Total Students," + total + "\n");
            fw.write("Male," + male + "\n");
            fw.write("Female," + female + "\n");
            fw.write(String.format("Male Percentage,%.2f%%\n", malePercent));
            fw.write(String.format("Female Percentage,%.2f%%\n", femalePercent));
            fw.close();
            System.out.println(fileName + " created successfully.");
        } catch (IOException e) {
            System.out.println(e.getMessage());
        }
    }
    public static void main(String[] args) {
        readCSV();
        processGroup("Group A", groupA);
        processGroup("Group B", groupB);
        processGroup("Group C", groupC);
        processGroup("Group D", groupD);
        System.out.println("\nAll group files created successfully.");
    }
}
```

OUTPUT:

```
Console
<terminated> Exercise7 [Java Application] C
CSV loaded successfully.

Group A
Total students: 81
Male: 50
Female: 31
Male percentage: 61.73%
Female percentage: 38.27%
groupa.csv created successfully.

Group B
Total students: 185
Male: 84
Female: 101
Male percentage: 45.41%
Female percentage: 54.59%
groupb.csv created successfully.

Group C
Total students: 308
Male: 134
Female: 174
Male percentage: 43.51%
Female percentage: 56.49%
groupc.csv created successfully.

Group D
Total students: 249
Male: 130
Female: 119
Male percentage: 52.21%
Female percentage: 47.79%
groupd.csv created successfully.

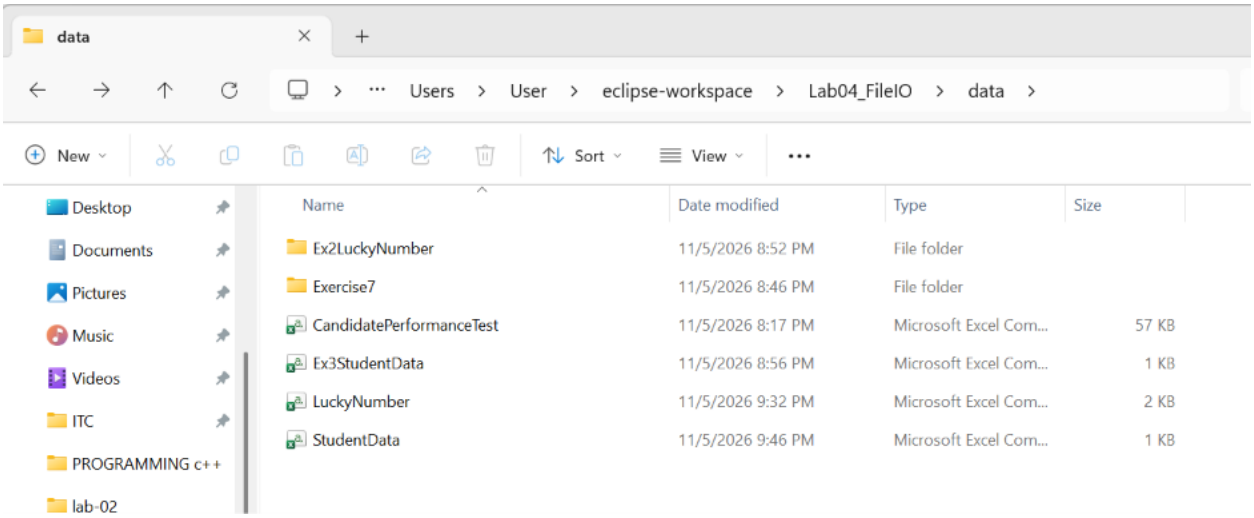
All group files created successfully.
```

URL GITHUB REPOSITY

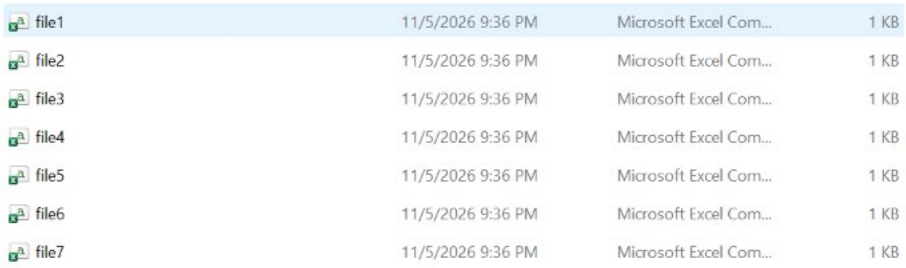
https://github.com/sethpanha902-arch/Lab04_FIleIO.git

FILE

Data folder:



Ex2LuckyNumber folder:



Exercise7 folder:

